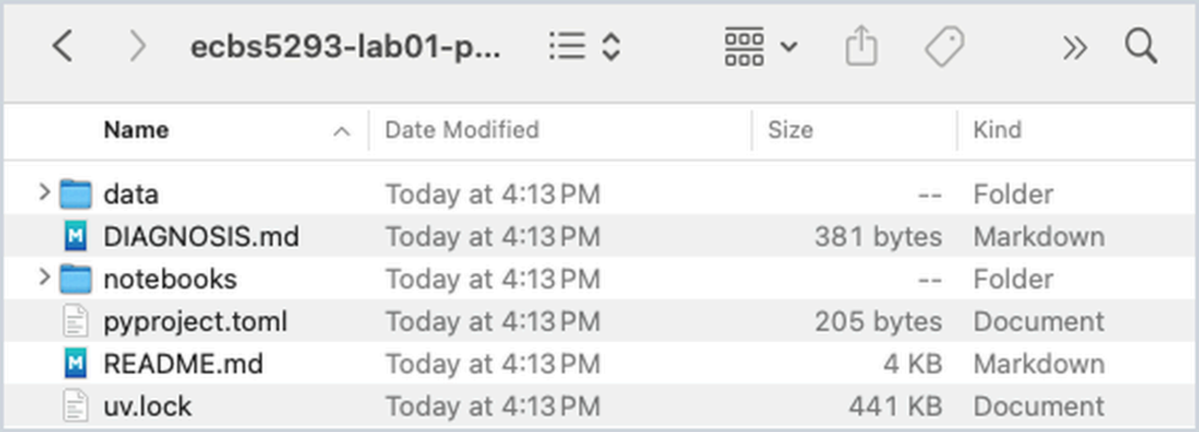
Computing for Analytical Work

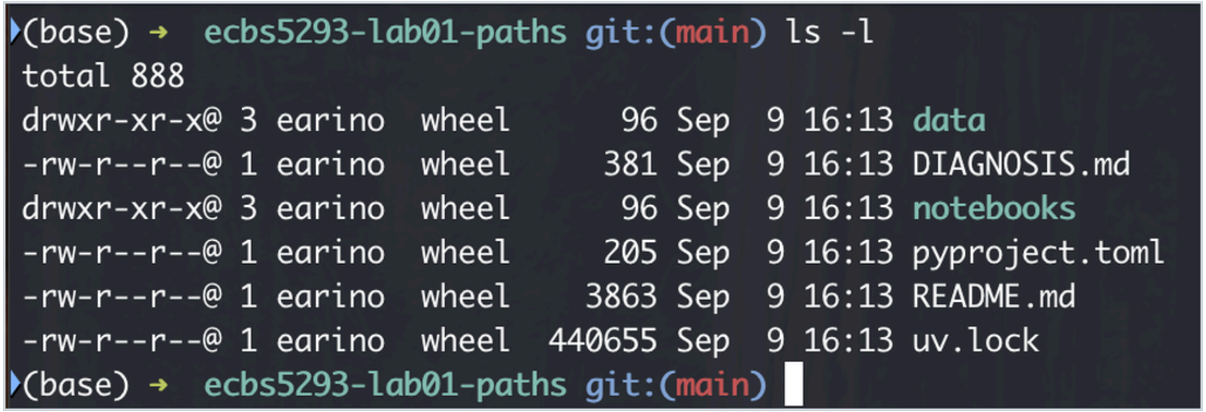
Session 1 — Where am I?

Block 2 — The terminal as a way to inspect reality

This block in one sentence

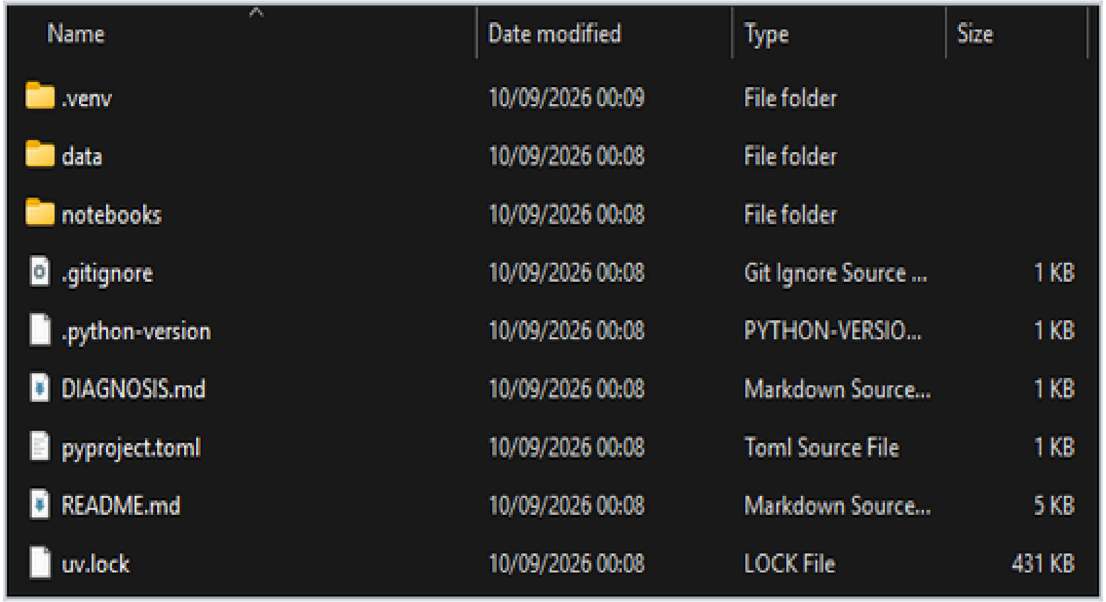
The terminal is not a different computer. It is a precise way to talk to the same computer.

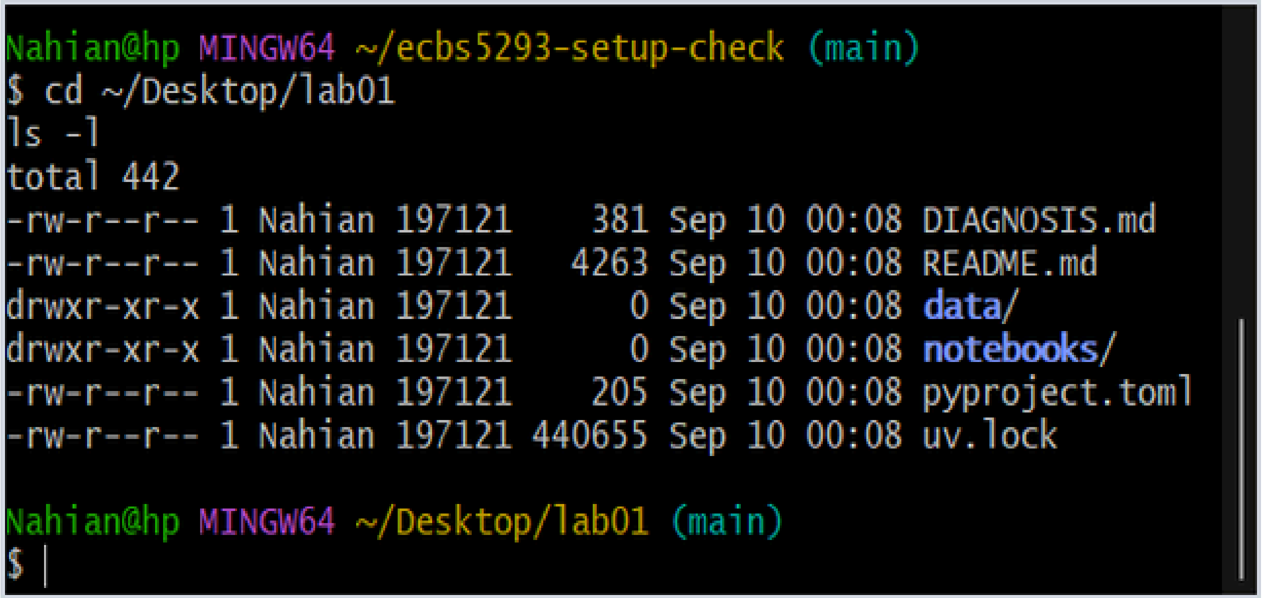




Same folder, same six entries, two views. Finder shows a tidy list. The terminal shows the same list plus exact sizes, times, owners, and permissions.

Same on Windows





File Explorer and Git Bash on the same folder. Explorer here is set to show hidden files, so it lists three more: `.venv`, `.gitignore`, `.python-version`. `ls` hides dot-files unless you ask: `ls -la`.

Agenda

1. Two-minute recap: `pwd`, `ls`, `cd`
2. Shell, prompt, command, output — and the Windows rule
3. Making and moving things: `mkdir`, `cp`, `mv`
4. Looking inside files without Excel: `head`, `tail`, `wc`
5. Running a script with `uv run python …`
6. Recording the commands that worked
7. Git thread: `git diff`. AI thread: ask before you run.
8. Stretch, Lab 2, Homework 1

Two-minute recap: `pwd`, `ls`, `cd`

```
pwd                # where am I?
ls                 # what is here?
cd notebooks       # go there; every relative path now starts from there
cd ..              # up one level
```

You used all four in Lab 1. If any of them still feels foreign, tell a TA during the lab. We are not re-teaching them; we are building on them.

What a shell is

- The **terminal** is the window. The **shell** is the program inside it that reads what you type.
- The shell shows a **prompt**, waits, you type a **command**, press Enter, it runs it, prints the **output**, and shows the prompt again.
- That loop is the whole thing. Prompt, command, output, repeat.

```
anna@laptop project $ ls data/raw
sales.csv
anna@laptop project $
```

Anatomy of a command

`ls` `-l` `data/raw`

1 2 3

1 **command** the program to run

2 **option (flag)** changes how it behaves: here, a long listing

3 **argument** what to act on: a path, resolved from the cwd

```
anna@laptop project $ ls -l data/raw
total 96
-rw-r--r--  1 anna  staff  48211 Sep  3 10:42 sales.csv
```

size in bytes last modified

evidence, before you open anything

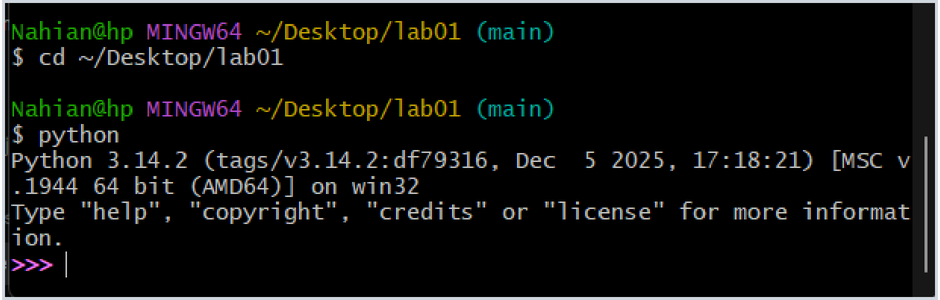
Spaces separate the pieces. A path with a space in it needs quotes: `"My Data/raw"`. Better: no spaces in project names.

Supported shells and the Windows rule

- **macOS / Linux**: the system Terminal. **Windows: Git Bash**, installed with Git for Windows. PowerShell and `cmd.exe` are at your own risk; TAs will not debug them in lab. Every command in this course works identically in both.

Windows rule: never type bare `python` in Git Bash. In the Git Bash window Git for Windows installs, it often hangs with no error. Inside VS Code's terminal it may just work. You cannot tell in advance, so always pass an argument:

```
python --version
python -c "import sys; print(sys.executable)"
```



If it *does* run and you see `>>>`, you are inside Python, not the shell. `exit()` gets you back.

Commands that change things

`mkdir`, `cp`, `mv`

`mkdir` — make a folder

```
mkdir output                # one folder, in the cwd
mkdir -p data/processed     # make parents as needed; no error if it exists
ls                          # confirm
```

- Relative to the cwd, like every other path.
- Programs do **not** create folders for you when they write files. A script that writes `output/summary.csv` into a missing `output/` fails. Remember this in the lab.

`cp` — copy

```
cp data/raw/sales.csv data/raw/sales_backup.csv   # file → new file
cp -r data data_backup                            # folder → folder, recursively
```

- Source first, destination second. Both relative to the cwd.
- Copying makes **two** files. Now there are two truths. Before you copy data, ask whether you actually want that.

`mv` — move, which is also rename

```
mv notebooks/sales.csv data/raw/sales.csv   # move to the right place
mv analyze.py analyse.py                    # same folder: that is a rename
```

- One file remains. That is the honest fix for a misplaced file.
- `mv` will **silently overwrite** a file at the destination. Look first: `ls data/raw`.

Predict: `mv` from the wrong place

You are standing in `project/notebooks/`. You type:

```
mv notebooks/sales.csv data/raw/sales.csv
```

What happens? Decide before the next slide. Hands: moves, or fails?

`mv` from the wrong place — what happened

```
mv: notebooks/sales.csv: No such file or directory
```

- Both arguments resolve from the cwd. The shell looked for `notebooks/notebooks/sales.csv`.
- Same rule as Block 1, now with a command that **changes** things.
- Had it succeeded from the wrong place, it would have said nothing. Look, then act.

Commands that only look

`head`, `tail`, `wc`

`head` — the first lines

```
head data/raw/sales_2024.csv          # first 10 lines
head -n 3 data/raw/sales_2024.csv    # first 3
```

```
anna@laptop project $ head -n 3 data/raw/sales_2024.csv
date,region,product,units,unit_price
2024-03-13,North,Doohickey,5,4.75
2024-06-19,North,Widget,33,12.5
```

1. **the header row**
   five names your code must match exactly
2. **the delimiter**
   a comma
3. **the date format**
   ISO: year-month-day
4. **the decimal mark**
   a point, not a comma

You now know the delimiter, the header, the date format, and the column names. Before writing a line of code.

`tail` — the last lines

```
tail data/raw/sales.csv          # last 10 lines
tail -n 2 data/raw/sales.csv     # last 2
```

- Files often go wrong at the end: a truncated download, a trailing "Total" row, a blank line, a footer someone typed by hand.
- `head` and `tail` sample the two **ends**. Together they tell you whether the ends agree. The middle stays unseen: `wc -l` is the next check, and a full pass comes in Session 3.

`wc` — how big is it?

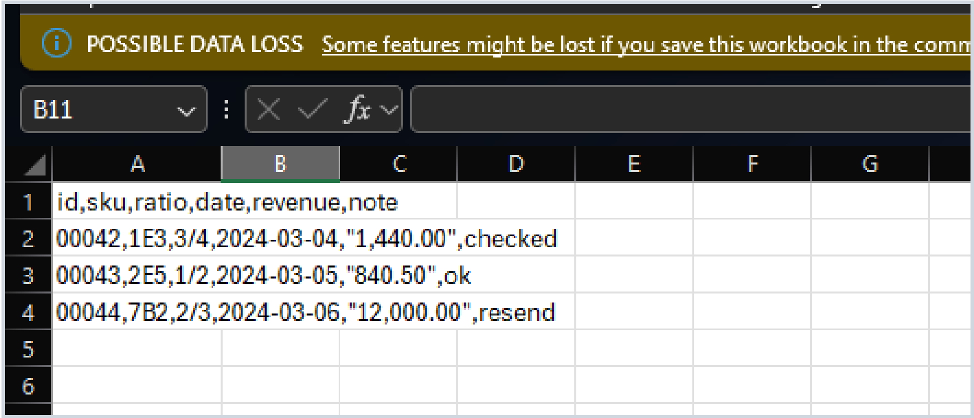
```
wc -l data/raw/sales.csv        # count lines
wc -l data/raw/*.csv            # all CSVs at once
```

```
    1201 data/raw/sales.csv
```

1,201 lines. If the header is one line, every record is one line, and nothing trails at the end: 1,200 rows. Say those assumptions when you subtract; a quoted field with a line break in it breaks the arithmetic. When your dataframe says 900, you have a number to argue with.

Why not just open it in Excel?

- Excel does not show you the file. It shows you its **interpretation** of the file.
- It reinterprets dates, drops leading zeros, reads `1,440.00` by locale, and depending on the Windows regional format may not split the columns at all: the whole row lands in one cell (right). On save it may change the delimiter or encoding.
- Then you click Save, and the file on disk is different from the one you were given.
- `head` shows the bytes. It cannot change them.



Rule for the course: inspect with the terminal; analyse in Python; open in Excel only when you mean to edit.

Predict: a script run from `scripts/`

Two launches. Inside the script, both times, the code opens `data/raw/sales.csv`.

```
cd scripts && uv run python analyze.py              # launch A
cd .. && uv run python scripts/analyze.py           # launch B, back at the project root
```

For each launch, resolve **two** paths from the shell's cwd, and decide whether each exists in the session tree:

1. The **script** path — the thing you typed after `python`.
2. The **data** path — the thing the script opens.

Decide, then the next slide.

Running a script

```
cd /path/to/project                     # the root — where the README says to be
uv run python scripts/analyze.py   # launch B: the script path and every path inside it resolve from here
```

- **Launch A**, from `scripts/`: the script path `analyze.py` resolves to `scripts/analyze.py`, exists. The data path resolves to `scripts/data/raw/sales.csv`, does not.
- **Launch B**, from the root: `scripts/analyze.py` exists, and the data path resolves to `data/raw/sales.csv`, exists. The cwd changed, so both paths changed meaning, and the command had to change to match.
- The script's cwd is **where you ran it from**, not where the `.py` file lives. Same rule as the notebook kernel in Block 1. Output goes wherever the script's relative paths point — from the cwd.

Sidebar: you have been typing `uv run` since setup

- The pre-course checklist had you run `uv sync` and `uv run python -c "import pandas; …"`.
- Lab 1 had you run `uv sync` and pick the `.venv` kernel. Lab 2 is `uv run python scripts/summarize.py`.
- For now it is a **recipe**: type it exactly, from the project root.
- Session 2, Block 3 opens by explaining what it has been doing all along. Until then, do not substitute bare `python`, and do not ask why. Just insist on it.

Recording the commands that worked

Keep a plain text file — `COMMANDS.md`, `notes.txt`, whatever — and paste in the exact lines that worked, in order, with the folder you were in.

```
# from the project root
uv sync
mkdir -p output
uv run python scripts/summarize.py
ls output
```

Not the ones you tried. The ones that **worked**. That file becomes your README's run instructions — and *those* are a homework deliverable.

Git thread — `git diff`: what did I change?

```
git status      # which files changed?
git diff        # what changed inside them?
```

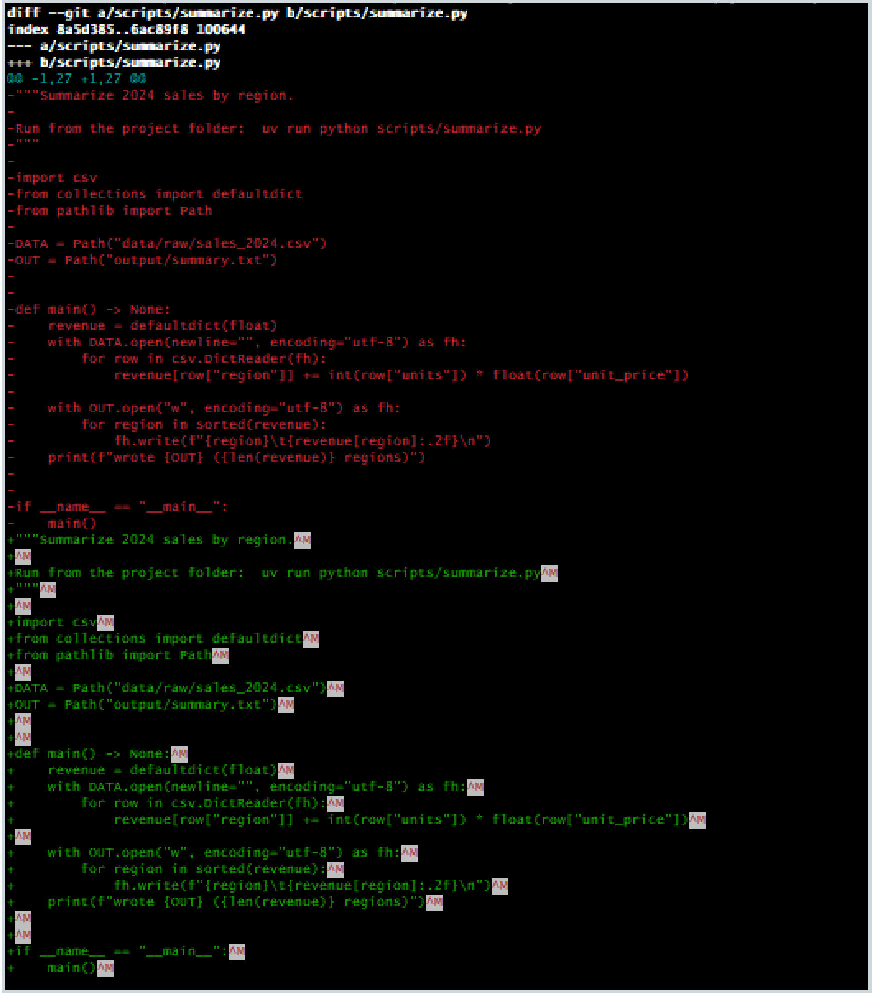
```
--- a/scripts/summarize.py
+++ b/scripts/summarize.py
-OUTPUT = "../output/summary.csv"
+OUTPUT = "output/summary.csv"
```

Lines starting with `-` are what was there. Lines with `+` are what you wrote. Everything else is context.

`git diff` covers **tracked** files only. A file you created is not in it. A file you *moved* shows as a deletion at the old place and nothing at the new one; `git status` lists the new location as *untracked*.

Git thread — why `git diff` matters today

- Before checkoff, run `git diff`. If the diff shows more than your fix, you changed more than you meant to.
- The diff **plus** `git status` are your "what did you change?" answer. Paste both into the diagnosis note; a move is a `deleted` line and an `untracked` line, so say it in words too.
- Windows: if `git diff` looks like the one on the right, every line removed and re-added with `^M` at the ends, `core.autocrlf` is not set to `input`. Fix it before you continue, or the diff is unreadable.



AI thread — ask before you run

```
I am in Git Bash on Windows, in the root of a Python project.
Before I run it, what does this command do, step by step?

    rm -rf output/*

Is there a version that only deletes CSV files, and asks before each one?
```

- **Explain before executing.** Especially anything with `rm`, `mv`, `-r`, `-f`, or `*`.
- **Ask for a safer version** of a destructive command. `rm -i`, a narrower pattern, a dry run.
- Then read the explanation, and decide.

AI thread — you are responsible for what runs on your machine

- The assistant did not press Enter. You did.
- "AI told me to" is not a diagnosis, not an excuse, and not something you can explain at checkoff.
- AI can help interpret commands and output. **You are responsible for what runs on your machine.**
- Good uses today: explain this command, explain this output, give me a safer version. Bad use: paste a block you cannot explain and run it.

5-minute stand-and-stretch

Stand up. Leave the laptop. Water, window, hallway.

Back in five minutes, then 45 minutes of lab. Homework 1 is on the last slide; it will still be there when you return.

Lab 2 — Run the project without clicking around

`scripts/summarize.py` reads a CSV with the standard library only — no pandas; this lab is about paths and the shell. First inspect the input with `head` and `wc -l`, as the README says. Then follow the README's *Reproduce the failure* block exactly and read what happens. Then the loop: observe → locate → hypothesize → inspect → change → verify. There is more than one thing wrong, and the first fix is not the last. Record the exact commands that worked. `git diff` to see your change. One diagnosis note per thing you fixed. Checkoff.

From the project folder, type this exactly, then the README's *Reproduce the failure* block:

```
uv sync
```

Stretch: make the script robust to the second thing you found, so the next person never meets it. And: why would this also run with bare `python scripts/summarize.py` and no `uv sync`? Hold that for Session 2.

Homework 1 assigned — Files, paths, terminal, and run instructions

A colleague's project that does not run as documented. Make it run, prove the report came from the supplied data, and document how. Expected time: **5–6 hours**. Due the **Friday after this session, 23:59** — slot on Moodle.

Deliverables, on Moodle:

1. The fixed repo with corrected paths. The grader regenerates the output from your run instructions — so show in your note that it did.
2. Run instructions in the README: the exact commands that worked, in order, from the project folder
3. A diagnosis note per failure: symptom, cause, evidence, change, verification — with the sanity counts (rows in, rows loaded, products out) **and the reconciliation**: revenue totalled from the input equals the report's total, to the cent
4. A **60–90 second video walkthrough**: screen recording, run the fixed code, then for every failure you fixed: what broke, why, what you changed, how you verified. No script reading.
5. The zip, made by one command in the README: `uv run python make_submission.py`. No Git needed yet — Git today is `status` and `diff`, and the diff belongs in your note.